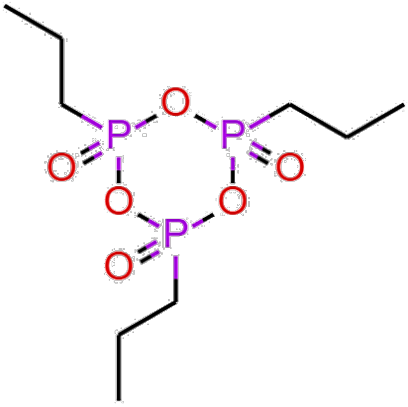


Genotek Biochem

ProPhos A3 (Propylphosphonic anhydride) CAS No. 68957-94-8

Reagent of choice for Green, Efficient Dehydro-Coupling!

Ultra-high purity & performance	Proprietary, eco-friendly process	Made-in-India
Atom-efficient technology for minimal waste process	Competitive Pricing & Quick Logistics	

ProPhos A3 (Propylphosphonic anhydride) is an exceptional reagent for amide/ peptide bond formation. It is very easy to use and has a high reaction selectivity, with almost no epimerization. It can also be used in esterification reactions. ProPhos A3 is a mild & efficient water removal agent with low toxicity and low allergenic potential.

ADVANTAGES

High Reactivity
Organic equivalent to hazardous reagents

Synthesis
Linear and cyclic oligoamides with thiophene backbone

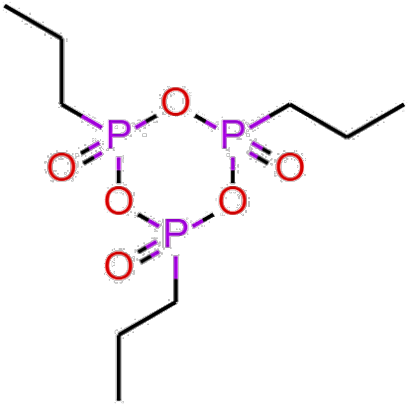
Better Scale-up
Low toxicity, low allergenic potential, non-flammability and broad functional group tolerance

Wider Solvent Range
Ethyl acetate, DMF, Acetonitrile, THF, Diethyl carbonate

Typical specifications for ProPhos A3:

Appearance	Colourless to Pale Yellow to Pale Brown liquid
Identification by NMR	Conforms
Assay by GC	NLT 50%
Active Content by Potentiometric Titration	NLT 45%
Store in tightly closed container under inert atmosphere in a cool & dry place. Highly moisture sensitive.	

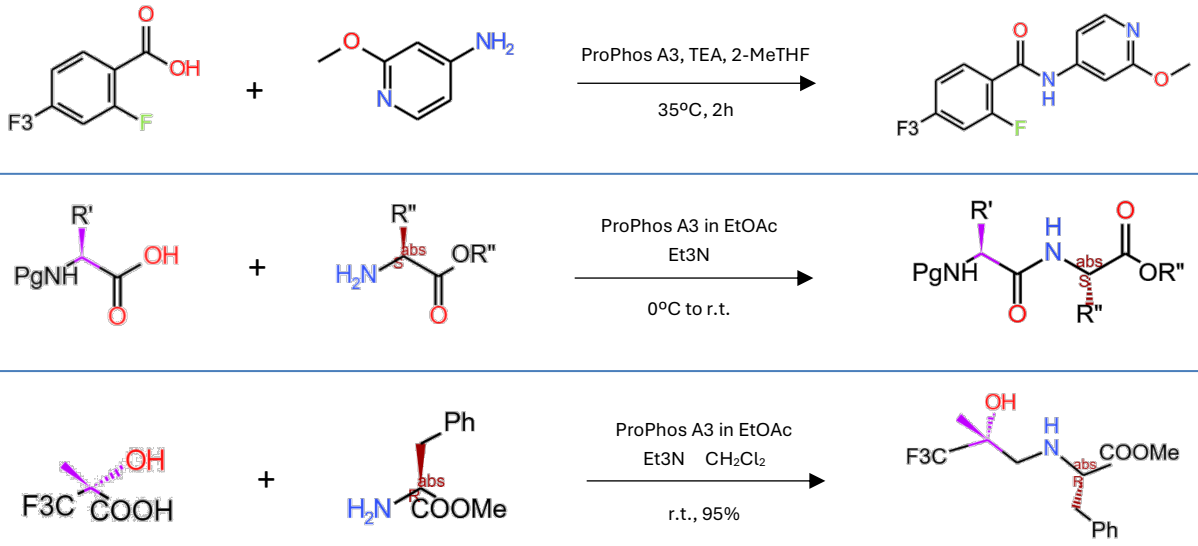
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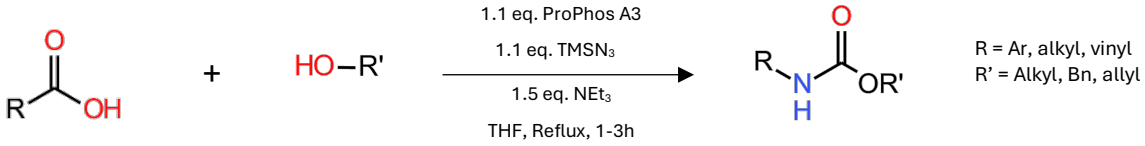
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Some established applications:

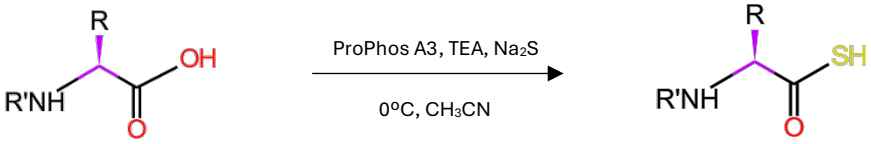
Peptide coupling reagent – higher yield and comparatively low epimerization, hence reducing the need of high column purification. By-products of the coupling reaction are water soluble and do not require special workup. It avoids additional step for converting acids to their more reactive derivative. Minimum racemization & without any additive to avoid racemization. N-methyl amino acids and α,α-di-substituted amino acids coupling with desired peptides in excellent yields and avoiding stericfactors.



Synthesis of Peptidomimetics and Amino acid derivatives – conversion of N-protected amino acids into the corresponding acidazides.



Synthesis of Amino and Peptide thioacids - N-protected amino or peptide thioacids starting from the corresponding amino or peptide acids, with good yield & optical purity.



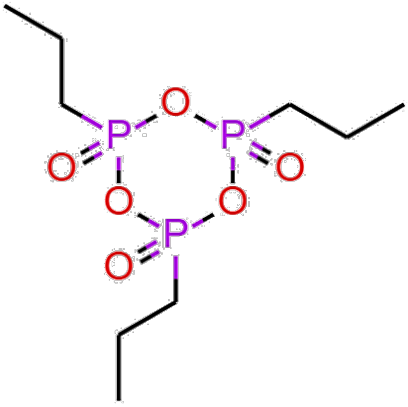
R' = Fmoc, Boc, Fmoc-Xaa, Z-Xaa, Boc-Xaa

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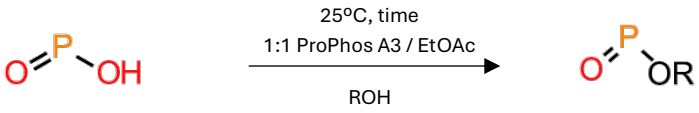
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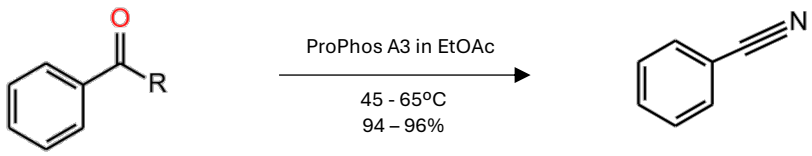
Synthesis of N-protected Amino acid esters – esterification of five membered cyclic phosphinic acid can be carried out under mild conditions.



R' = Me, Et, Pr, 'Pr, Bu, ^{5}Bu, Pent, ^{1}Pent, 3-Pentyl, Hexyl, Bn, 2-Phenylethyl, 2-(1-Naphthyl)ethyl, Menthyl

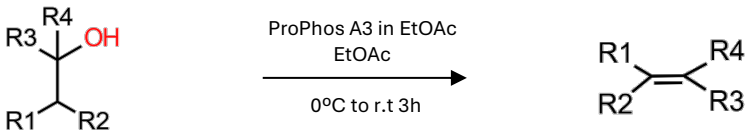
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Preparation of Nitriles – convert acids, ammonium salts and amides into nitriles. Other products formed in process are water soluble and easy for isolation.



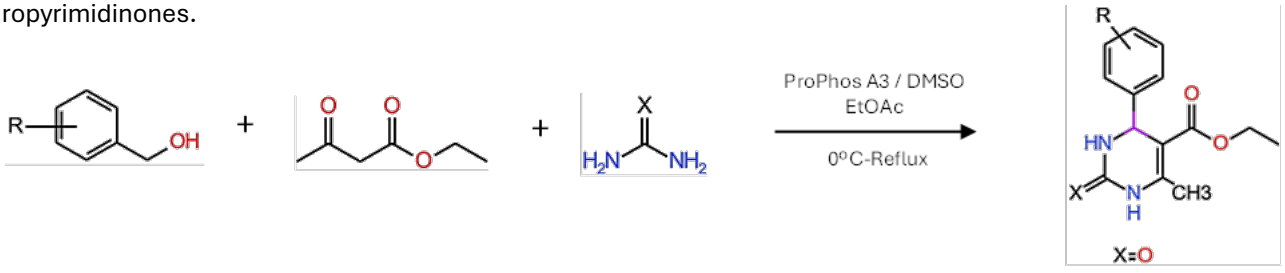
R = OH, ONH_4, NH_2

Preparation of Alkenes from Alcohols – conversion with primary, secondary & tertiary alcohols with very mild & simple conditions.

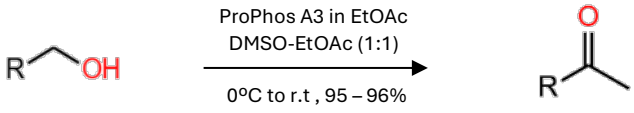


R^1, R^2 = Alkyl, R^3, R^4 = H
R^1, R^2, R^3 = Alkyl R^4 = H
R^1, R^2, R^3, R^4 = Alkyl

Formation of Biologically Active Substituted Heterocycles – synthesis of biologically and pharmaceutically active heterocycles like Oxa & Thiadiazoles, Quinolines, β-Lactams, Indole, Benzothiazoles, Naphthyridines, Coumarins and Dihydropyrimidinones.



Selective Oxidation of Alcohols to Carbonyl Groups – selective oxidation of alcohols to carbonyl, which are used in preparation of fine & specialty chemicals.



R = Ph, CH_2=CH_2, $Me(CH_2)_7$, Bn

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